Final Exam : Find the Missing Number

CSE 130: Algorithm Design

Steve is a teacher and has given each of his students an ID number from 1 to N where there are N students in his class. Every day, he has his students stand in line and turn in their homework according to their ID number. On occasion, Steve notices that a student is missing. The problem is, which one…

Assignment Definition

You are given an array of size *n* with the range of numbers from 1 to *n*+1. This array has no duplicates, one number is missing, and the array is not sorted. Find the missing number. Of course, we want this design to be as efficient as possible…

Please do the following:

- Create a representation of a solution to this problem using any combination of design tools presented this semester
- Evaluate the quality of your design using appropriate metrics presented this semester
- Validate the design by applying appropriate quality tools or techniques

Grading

	Exceptional *100%*	*Good* *90%*	*Acceptable* *70%*	*Developing* *50%*	*Missing* *0%*
Design Quality 60%	The optimal solution was found	A workable solution was found that has acceptable performance characteristics	A correct solution was found with poor performance or a minor flaw exists with the presented solution	Part of the problem was solved but there exists serious issues with the design	The solution as presented is not on track to solve the problem
Tools 10%	The design was unambiguously and clearly described	An appropriate design tool was utilized and the tool was used without error	Minor ambiguity exists in the design presentation or a minor error exists in how the design tool was utilized	The design tool was utilized poorly	The solution does not demonstrate mastery of the design tools of the semester
Metrics 10%	Every nuance of the design was correctly characterized	A suitable metric was utilized and the metric was used without error	A minor error exists in the use of the metric	A serious problem exists in the use of the quality metric	No knowledge of a quality metric was demonstrated
Quality 20%	It is difficult to image how a defect could exist in the code	The design was verified through a suitable application of a quality technique	The design was not adequately verified, but knowledge of quality techniques were demonstrate	Familiarity with quality techniques were demonstrate somewhere in the submitted paper	No knowledge of quality techniques were demonstrated